Here is a comprehensive, structured breakdown of **all differences** between the local (reference) version and the online version, organised by category: --- ## 🔴 PAGE HEADER / SESSION IDENTITY | Element | LOCAL (correct) | ONLINE (needs fixing) | |---|---|---| | **Page title** | OK_R3 | /TMP/TELEMETRY_UPLOAD_YSE1WGRP/OK_R3.CSV (raw upload path shown instead of session name) | | **Driver name** | Driver_1 | Théo (different driver name — likely a metadata mismatch) | | **Track name** | Ouest Karting (resolved, named track) | Unknown Track (not resolved) | | **Date** | 02/28/2026 (session date) | 2026-03-01 (wrong date — showing upload date instead of session date) | | **Subtitle line** | Not displayed as a subtitle | Shows Théo • Unknown Track • 2026-03-01 as a subtitle — all three values are wrong | | **Breadcrumb** | ← Sessions (simple back link) | Sessions / (slash separator, different style) | --- ## 🔴 SESSION INFO PANEL (top metadata block) | Element | LOCAL | ONLINE | |---|---|---| | **Editable fields** | Date, Driver, Track, Type, Condition, Kart are all **editable dropdowns/inputs** | SESSION INFO panel is **read-only**, no editing capability | | **Kart field** | Shows Sodikart Sodi RT10 (GX390) — kart linked | **No kart field at all** in the online Session Info panel | | **Best Lap** | Shown as a stat card (74.496s in orange) | Not shown in header area | | **Mean Lap** | Shown as a stat card (77.537s) | Not shown in header area | | **Score** | Shown as a stat card (71.9 in orange) | Not shown in header area (only shown inside Overview tab) | | **Score label** | Shows "Conditions" (field label) | Online shows "Conditions" ✓ — but "Condition" in local vs "Conditions" online (minor label difference) | --- ## 🔴 NAVIGATION / TAB STRUCTURE | Element | LOCAL | ONLINE | |---|---|---| | **Tab names** | Overview / Corners & Coaching / Timeline / Analytics / Lap Data | Overview / Visualize / Corners / Performance / Lap Data | | **Tab: "Corners & Coaching"** | Full tab with animated replay, coaching table, heatmaps, key corners | Online calls it Corners and shows **"No corner coaching data available"** — entirely empty | | **Tab: "Timeline"** | Full timeline with lap-by-lap events, PBs, hot laps, visual timeline | **Missing entirely** online (replaced by "Visualize" and "Performance" tabs) | | **Tab: "Analytics"** | Full tab: Track Map, Sector Performance, Lap Speed Comparison, Lap Distribution, Lap Progress Tracker, Cross-Session comparison, G-G Explorer, Suggested Drills | **Missing entirely** online | | **Tab: "Visualize"** | Part of the Analytics tab (Track Map + Speed Trace) | Exists as a standalone tab online — but maps are **empty/blank**, no data rendered | | **Tab: "Performance"** | Exists as part of Overview tab content | Online has it as separate tab with basic metric cards only (no full summary, no charts) | | **Focus Mode button** | Present (⊞ icon, top-right of tab bar) | **Missing** | | **Tab active color** | Orange underline | Green/lime underline (different color scheme for active tab indicator) | --- ## 🔴 OVERVIEW TAB CONTENT | Element | LOCAL | ONLINE | |---|---|---| | **Score Breakdown subtitle** | "Your overall performance scored across four pillars. Hover each label for a breakdown..." | No subtitle text | | **Score max values (denominators)** | Rounded: 21 / 26 / 17 / 20 (clean integers) | Raw decimals: 21.028 / 25.7009 / 16.8224 / 19.6262 (unformatted) | | **Adaptation score** | 11.2 (no denominator shown) | 11.3 / 15 (different value AND shows the denominator) | | **Strengths section** | Present — lists: Pace, Execution Quality, Racecraft / Endurance Robustness | **Missing** | | **Areas to Improve section** | Present — lists: Consistency, Driving Style | **Missing** | | **Driving Style section** | Full section: style label ("Aggressive Attack"), confidence %, 9 trait bars (Trail Braking, Brake Consistency, Brake Attack, Apex Speed, etc.), Style Strengths & Weaknesses lists, radar chart | **Missing entirely** | | **Session Debrief section** | Full AI-generated debrief: Grade card, highlights, key metrics table, areas for

improvement, next session actions | **Missing entirely** | | **Achievements & Rating section** | Full badges section: 4 badges (Pace Setter ★Gold, Flow Master ★Bronze, Quick Learner ★Gold, Iron Will ★Gold), star rating, badge insights | **Missing entirely** | | **Right-side Session Info panel** | Not present in local (info is in the editable top fields) | Present online as a static panel — partially replaces the editable header | --- ## 🔴 CORNERS & COACHING TAB | Element | LOCAL | ONLINE | |---|---|---| | **Key Corners Replay** | Animated cockpit view with speedometer, corner navigation (P1, P2, P6), play/pause, speed buttons (0.25x, 0.5x, 1x, 2x), Corner Review / Split Comparison / Top 3 Overlay modes | **Not rendered / missing** — tab shows "No corner coaching data available" | | **Corner Coaching table** | Full table: C1–C5, Entry/Apex/Exit speeds, Grip %, Loss (ms), Coaching hints | **Missing** | | **Key Corners checkboxes** | Interactive checkboxes per corner (C1–C5) with ms loss labels, Save button | **Missing** | | **Brake Precision chart** | Present (shows "no data" placeholder, but section exists) | **Missing** | | **Throttle Onset Map** | Present (placeholder) | **Missing** | | **Consistency Heatmap** | Present (placeholder) | **Missing** | | **Corner Mastery Report Card** | Present (loading placeholder) | **Missing** | --- ## 🔴 ANALYTICS TAB (local) — ENTIRELY MISSING ONLINE The entire **Analytics** tab doesn't exist online. The following sections are lost: - **Track Map** (interactive, with GPS overlay, lap selector, all-laps toggle, corner markers C1/C2/C4/C5, zoom, speed trace, primary/compare lap overlays) - **Sector Performance** - **Lap Speed Comparison** (multi-lap overlay chart with checkboxes) - **Lap Time Distribution** - **Lap Progress Tracker** - **Session Comparison Matrix** - **Cross-Session Comparison** (percentile, vs last session, comparable sessions count) - **Cross-Session Speed Overlay** (multi-session chart) - **G-G Explorer** (All/Single/Compare modes) - **Suggested Development Drills** (Consistency Focus, Pace Extraction with drill details) --- ## 🔴 LAP DATA TAB | Element | LOCAL | ONLINE | |---|---|---| | **Performance Summary block** | 8 stat cards: Best Lap, Mean Lap, Std Deviation, Consistency %, Best vs Mean, Top 25% Avg, Improvement, Valid Laps | **Missing entirely** online | | **Lap Times chart** | Bar chart with mean line, shows laps 1–9, best highlighted in orange, incomplete lap with dashed pattern | **Missing** online | | **Lap Data table columns** | LAP / TIME (s) / AVG SPEED / MAX SPEED / MAX BRAKING (G) / MAX ACCEL (G) / LINE (m) / PHASE | Online: LAP / TIME / AVG SPEED / MAX SPEED / **STATUS** only — 3 columns missing (Max Braking, Max Accel, Line) | | **Time format** | Decimal seconds: 82.031, 81.710 etc. | MM:SS.mmm format: 1:22.031, 1:21.710 — different format | | **Speed units label** | No inline unit (implied km/h) | Inline label KM/H per cell | | **Lap 8 "Best" badge** | Orange Best badge on row 8 | No badge — just shows Valid like all other laps | | **Lap 9 "Incomplete" badge** | Amber Incomplete badge, row dimmed | Shows Valid — incorrect status | | **Track Condition Phases** | Section present: Phase 0, Laps 1–9, Conf: 100% | **Missing** | --- ## 🟡 MINOR / STYLE DIFFERENCES | Element | LOCAL | ONLINE | |---|---|---| | **Top nav structure** | Horizontal top bar with dropdown menus (Data, Analysis, Visualize, More) | Vertical left sidebar with category sections | | **App name/logo** | Telemetry (logo + wordmark) | LAP (different branding) | | **Active tab underline color** | Orange | Lime/green | | **Session heading style** | Large OK_R3 with orange underline decoration | Raw file path in all-caps, smaller styling | | **Score bar colors** | Pace=green, Consistency=red/salmon, Execution=green, Robustness=green, Adaptation=blue-purple gradient | Pace=lime, Consistency=cyan, Execution=magenta/pink, Robustness=orange, Adaptation=yellow-gold | | **"Download Report" button** | Not present in

local | Present online (top-right) | | **Language selector** | EN dropdown (with FR/IT/DE/ES/PT/ZH options) | Not present | | **Scroll-to-top button** | Present (orange ↑ circle) | Not visible | | **Footer** | Telemetry Analysis Platform | No footer | --- ## Summary of Priority Fixes **Critical (major missing functionality):** 1. Session metadata wrong (name, driver, track, date) 2. Entire **Timeline** tab missing 3. Entire **Analytics** tab missing (largest tab — 10+ sections) 4. **Corners tab** empty — all coaching data missing 5. **Overview** missing Driving Style, Session Debrief, Achievements sections 6. **Lap Data** missing Performance Summary, Lap Times chart, 3 table columns, correct badges, Track Condition Phases **High (data/format issues):** 7. Score denominators showing raw decimals instead of rounded integers 8. Adaptation score value differs (11.2 vs 11.3) 9. Lap 9 shown as "Valid" instead of "Incomplete" 10. Time format (decimal seconds vs MM:SS) **Medium (UI/UX):** 11. No editable session fields online 12. No Focus Mode buttons 13. Different color scheme for score bars 14. Active tab color mismatch (orange vs green) Here is the complete diff for every other section of the app beyond the session report: --- ## 1. DASHBOARD | Element | LOCAL (correct) | ONLINE (needs fixing) | |---|---|---| | **Page subtitle** | None (clean heading only) | Shows subtitle "Your telemetry overview" below heading | | **Stat cards** | Sessions, Drivers, Tracks, Total Laps, **Best Score**, **Avg Score** — 6 cards | Sessions, Drivers, Tracks, Total Laps, **Personal Best** (new), Best Score, Avg Score — 7 cards. "Personal Best" is a new card not in local | | **Personal Best card** | Not present | Added: shows 0:44.102 — this card doesn't exist in local | | **Score / Avg format** | 79.0 and 75.2 (plain number) | 85.5 /100 and 77.1 /100 (shows "/100" as a separate element) | | **Score Trend chart** | In "Overview" section at bottom, together with Best Lap Times | Shown **above** the Recent Sessions table, at top of dashboard | | **Lap Time Trend chart** | Called "Best Lap Times" in "Overview" section | Called "Lap Time Trend" with subtitle "Last 20 sessions" | | **Score Trend subtitle** | No subtitle | "Last 20 sessions" subtitle | | **Overview section** | Exists as a named section with heading "Overview" containing both charts | Charts are inline in the page body, no section heading | | **Recent Sessions table — columns** | Date / Driver / Track / Type / Laps / Best Lap / Score / Grade | Date / Track / Driver / Laps / **Best Time** / Score / Grade — column order changed, "Type" missing, "Best Lap" renamed to "Best Time" | | **Recent Sessions table — column headers sortable** | Static text headers | **Sortable buttons** (clickable column headers) | | **Recent Sessions table — rows as links** | Each row is a <a> link to the session | Each row is a <button> — different interactive pattern | | **Recent Sessions count** | Shows last 4 sessions | Shows last 6 sessions | | **"View All" button position** | Top-right of Recent Sessions section | Becomes "View All Sessions" button below the charts area | | **"Compare Sessions" shortcut** | Not present on dashboard | New "Compare Sessions" quick-link button added | | **"+ New Analysis" button** | Top-right of page next to heading | Below the charts, before the table | | **Scroll-to-top button** | Present (orange circle) | Not visible | | **Footer** | "Telemetry Analysis Platform" | No footer | --- ## 2. SESSIONS LIST PAGE | Element | LOCAL | ONLINE | |---|---|---| | **Page title** | Sessions with count badge 4 | Sessions (no visible count at list level — data shown in Visualization page instead) | | **Bulk action buttons** | Re-Analyze Selected + Delete Selected buttons in header | Not present | | **Table columns** | Date / Driver / Track / Type / **Condition** / Laps / Best Lap / **Mean Lap** / Score / Grade — 10 cols | Date / Track / Driver / Laps / Best Time / Score / Grade — 7 cols; **Condition**, **Mean Lap**, and **Type** columns missing | | **Column order** | Driver before Track | Track before Driver | | **Lap count label** | "Laps" column shows

lap count | Same ✓ | | **Time format** | 70.462s (decimal seconds) | 0:58.406 (MM:SS format) | | **Session rows** | Linked rows navigate to session | Rows appear as cards in Visualization, not traditional table rows | | **Search/filter bar** | Not present | Visualization page has a "Search by driver, track, date..." bar | | **Sort controls** | Not present | Visualization page has "Visualize →" per card and summary stats | --- ## 3. ANALYZE / UPLOAD PAGE | Element | LOCAL | ONLINE | |---|---|---| | **Page title** | New Analysis | Upload Session | | **Upload flow** | 3-step wizard: **① Select Data → ② Configure → ③ Analyze** with numbered progress indicators | Single flat form — no step wizard | | **File browser** | Shows data/raw/ directory listing with filenames + sizes | Simple drag-and-drop area only | | **Existing files list** | Shows OK_R4.csv (57.8 MB, 2026-03-01 05:07) as a selectable file | No file browsing — upload only | | **"Next: Configure" button** | Present (step-based) | Replaced by single **"Upload & Analyze"** button | | **Driver Name field** | Not on Step 1 (comes in Configure step) | Required text field present on same page | | **Session Type field** | Not on Step 1 | Dropdown present (Practice/Race/etc.) | | **Track Condition field** | Not on Step 1 | Dropdown present (Dry/Wet/etc.) | | **Lap Range field (optional)** | Not visible on Step 1 | Optional text field present | | **Exclude Laps field** | Not visible on Step 1 | Optional text field present | | **Button color** | Orange | Lime/green | | **Cancel button** | Not present | "Cancel" link back to sessions | | **Navigation breadcrumb** | "+ Analyze" top-right button in navbar | "Back to Sessions" text link top-right | --- ## 4. COMPARE PAGE | Element | LOCAL | ONLINE | |---|---|---| | **Page title** | Head-to-Head Comparison | Compare | | **Page subtitle** | None | "Side-by-side analysis of sessions and drivers" | | **Layout** | Full **interactive form** — two panels side by side, each with dropdowns pre-populated with real session/driver data + action buttons | **Hub/landing page** — two clickable cards that link to sub-pages (/compare/sessions and /compare/drivers). No dropdowns visible on this page | | **Session A & B dropdowns** | Present and pre-populated with 4 sessions each (with date, driver, track, best lap) | Not on this page — behind a click-through | | **Driver A & B dropdowns** | Present and pre-populated with 4 drivers | Not on this page — behind a click-through | | **"Compare Sessions" button** | Orange action button | "Open →" text link | | **"Compare Drivers" button** | Orange action button | "Open →" text link | | **Immediate usability** | Can select and compare directly from this page | Requires an extra click to reach the actual comparison form | --- ## 5. TRACKS PAGE | Element | LOCAL | ONLINE | |---|---|---| | **Layout** | Grid of track **cards** (each with name, sessions count, best lap, map preview indicator) | Summary stat cards at top (Total Tracks, Total Sessions, Avg Laps, Unique Drivers) + search bar + sort dropdown + card grid | | **Summary stat cards** | Not present | 4 stat cards: Total Tracks, Total Sessions, Avg Laps, Unique Drivers | | **Search bar** | Not present | "Search tracks, circuits..." search input | | **Sort dropdown** | Not present | "Most Sessions" sort dropdown | | **Track card data (local)** | Shows track name, sessions count, best lap | Online shows same general idea but data is empty (0 tracks shown due to no synced data) | | **Scroll-to-top / footer** | Present | Not present | --- ## 6. DRIVERS PAGE | Element | LOCAL | ONLINE | |---|---|---| | **Data shown** | 4 driver cards: 1 (0 sessions), Driver_1 (2 sessions, PB 70.462s, Score 74.6, Style: Adaptive Intuitive), TestDriver (1 session, Style: Unknown), Unknown (1 session, Style: Adaptive Intuitive) | 0 drivers — empty state, "NO DRIVERS YET" with icon | | **Layout (local)** | Simple grid of driver cards (name, sessions, PB, avg score, style) | 4 stat cards at top (Total Drivers, Total Sessions, Total Laps, Top Avg Score) + search bar + sort dropdown + empty card area | | **Summary stat cards** | Not present | Present (all show 0)

| | **Search bar** | Not present | Present: "Search drivers..." | | **Sort dropdown** | Not present | Present: "Most Sessions" | | **Driver card fields** | Name / Sessions / Personal Best / Avg Score / Style | Same structure but empty | | **Style field on cards** | Shows calculated driving style (e.g., "Adaptive Intuitive") | Would show same if data existed | | **Empty state** | Not applicable (has data) | Shows icon + "NO DRIVERS YET" + instructional text | --- ## 7. KARTS PAGE | Element | LOCAL | ONLINE | |---|---|---| | **Data shown** | 5 kart cards: Sodikart Sodi RT10 (GX270/GX390), Sodi SPORT (Rotax 28hp), SR5 (GX270/GX390) — with model, engine, power, sessions | 0 karts — empty state, "NO KARTS YET" | | **"+ New Kart" button** | Present (orange, top-right) | Present (orange, top-right) ✓ | | **Summary stat cards** | Not present | 4 stat cards: Total Karts, Total Sessions, **Avg Power (HP)**, **Avg Weight (KG)** | | **Avg Power / Avg Weight cards** | Not present | New cards online | | **Search bar** | Not present | "Search karts, chassis, engines..." | | **Sort dropdown** | Not present | "Most Sessions" sort dropdown | | **Kart card fields (local)** | Model / Engine / Power / Sessions | Same fields | | **Empty state** | N/A (has 5 karts) | Icon + "NO KARTS YET" + "Add your karts to track chassis, engine, and weight configurations" + **"ADD FIRST KART"** CTA button (extra button not in local) | --- ## 8. WEEKENDS PAGE | Element | LOCAL | ONLINE | |---|---|---| | **Data shown** | 1 weekend card: "Jovikart Cup Manche 1" — Track: Ouest Karting, Dates: 2026-01-25, Sessions: 0 | **Error state** — "Failed to load / Couldn't load weekends." with RETRY button | | **"+ New Weekend" button** | Present (orange) | Present (orange) ✓ | | **Summary stat cards** | Not present | 4 stat cards: Total Weekends, Total Sessions, Track Days, Unique Tracks | | **Search bar** | Not present | "Search weekends, tracks..." | | **Sort dropdown** | Not present | "Most Recent" sort dropdown | | **Weekend card fields** | Name / Track / Dates / Sessions | Same concept but page errors before rendering | | **Error handling** | No errors | Shows "Failed to load" — API error, page broken | --- ## 9. LEADERBOARD PAGE | Element | LOCAL | ONLINE | |---|---|---| | **Status** | Fully functional | **"Re-port in progress. This page is being rebuilt."** — completely empty/placeholder | | **Content** | Full leaderboard table: # / Driver / Best Lap / Avg Score / Best Score / Sessions / Laps / Style / Trend columns. 3 drivers shown with gold/silver/bronze medal icons, Stable trend indicators | Nothing — just the placeholder message | | **Filter controls** | Track dropdown ("All Tracks") + Sort dropdown ("Best Lap") | None | | **Medal/rank badges** | Gold, silver, bronze medal emoji icons per row | Not present | | **Style badges** | Purple pill badge ("Adaptive Intuitive") per driver | Not present | | **Trend column** | Shows "→ Stable" indicator | Not present | --- ## 10. ANALYTICS SUB-PAGES ### Weather & Pace (local) vs Weather (online) | Element | LOCAL | ONLINE | |---|---|---| | **Status** | Functional: "Weather & Pace Correlation" page with Track + Driver dropdowns and Analyze button, chart area | **"Re-port in progress. This page is being rebuilt."** — blank | | **Page title** | Weather & Pace Correlation | Weather (just a title, no content) | | **Breadcrumb** | ← Analytics back link | None | ### Pace Trends (local) vs Trends (online) | Element | LOCAL | ONLINE | |---|---|---| | **Status** | Functional: "Pace Trend Analysis" with Driver + Track dropdowns and Analyze button | **"Re-port in progress. This page is being rebuilt."** — blank | ### Live Meteo (local) — equivalent missing online | Element | LOCAL | ONLINE | |---|---|---| | **Status** | Fully built: "Live Meteo Dashboard" with Select Track dropdown, "Get Current Weather" button, 12h Forecast tab, Auto-refresh toggle + interval dropdown, Historical Weather date input + "Look Up Past Weather" button | **Not present at all** — no equivalent route exists online | ### Analytics Hub (online only) — new addition | Element | LOCAL | ONLINE | |---|---|---| | **Status** | Not present

— accessed as individual pages under Analysis menu | **New hub page** at /app/analytics with 5 feature cards: Weather Analysis, Performance Trends, Meteo Correlation, Weight Simulator, Pace Predictor — each with "Explore →" link | | **Pace Predictor** | Not present in local at all | New card in online hub (though may also be behind a placeholder) | | **Weight Simulator card** | Under Analysis > Weight Simulator directly | Accessible via hub card | --- ## 11. WEIGHT SIMULATOR | Element | LOCAL | ONLINE | |---|---|---| | **Status** | Fully functional: full form with Baseline Parameters, Load from Driver Profile, Best Lap Time input, Current Weight input, Vehicle Class dropdown, Weight Change slider (-30 to +30 kg), "Simulate Weight Change" button, Multi-Scenario Comparison section | Accessible via Analytics Hub card → but routes to same-looking page. **Needs verification** — hub link says "Explore →" and may also be a placeholder | | **Breadcrumb** | ← Setup Tracker | Not visible | --- ## 12. RACE STRATEGY PAGE | Element | LOCAL | ONLINE | |---|---|---| | **Data shown** | Empty: "No races yet. Create your first race to get started." | **5 race cards** visible: "TEST" (3h, 4 stints), "UPDATED TEST RACE (COPY)" (1h, 2 stints), "E2E TEST RACE (COPY)" (1h), "E2E TEST RACE" (1h), "TEST 4H ENDURANCE" (4h, 4 stints) — all FINISHED | | **Page title style** | Race Strategy (sentence case) | RACE STRATEGY (all caps) | | **Summary stat cards** | Not present | 4 stat cards: Total Races (5), Live Now (0), Completed (5), Race Hours (10.0h) | | **Search bar** | Not present | "Search races, tracks, teams..." | | **Status filter** | Not present | "All Status" dropdown | | **Import button** | Choose File / Import (file picker + Import button) | Not visible — import not on this page | | **Race card layout** | N/A (no data) | Cards show: name, status badge, Duration / Stints / Created / Action (SUMMARY →) | | **"+ New Race" button** | Present (orange) | Present (orange) ✓ | --- ## 13. COACHING / DEVELOPMENT PLANS PAGE | Element | LOCAL | ONLINE | |---|---|---| | **Page title** | Driver Development Plans | Development Plans (shorter) | | **Content** | Rich content: "Select Driver" dropdown + "Create New Plan" section with 4 plan template cards (Beginner Foundations, Intermediate Refinement, Advanced Performance, Race Weekend Preparation), each with description, level tags, duration (4 weeks), drill count (12 drills) | Empty: "0 plans / No development plans found." | | **Plan templates shown** | 4 plans visible as cards with full descriptions and tag pills | None | | **"Select Driver" section** | Present with driver dropdown | Not present | | **"Create New Plan" section** | Present with plan template grid | Not present | | **Empty state** | Not triggered (has content) | Triggered — blank with just the message | --- ## 14. VISUALIZATION PAGE | Element | LOCAL | ONLINE | |---|---|---| | **Route** | Accessed via Visualize dropdown in nav with inline session search, not a dedicated page | Dedicated page at /app/visualize | | **Layout** | Dropdown in nav bar with search box — no standalone page | Full page with summary stats + session card grid | | **Summary stat cards** | Not present | 4 stat cards: Recent Sessions (6), Total Laps (52), Avg Score (77.1/100), Scored (4) | | **Session cards** | Not a standalone page | Grid of 6 session cards, each showing: date, type badge, driver name, track, Laps/Best/Score/Grade, "Visualize →" button | | **Search bar** | Small inline search in nav dropdown | Full-width "Search by driver, track, date..." bar | | **Time format** | Would show decimal seconds | Shows MM:SS format (e.g., 0:58.406, 1:14.496) | --- ## 15. EXPORTS PAGE (ONLINE ONLY — missing locally) | Element | LOCAL | ONLINE | |---|---|---| | **Exists** | **Not present** — no equivalent page | Present at /app/exports | | **Content** | — | "Generate Export" form: Data Type dropdown (All Sessions) + Format toggle buttons (CSV / JSON / PDF / PARQUET) + "Generate" button. Export history area: "No exports yet." | | **This is a new

feature** | Not in local at all | Entirely new page in online version | --- ## SUMMARY TABLE — Page-by-Page Status | Page | LOCAL | ONLINE | Status | |---|---|---|---| | Dashboard | ✅ Full | ⚠️ Modified layout, extra cards, wrong column order | Regression | | Sessions List | ✅ Full | ⚠️ Missing columns (Condition, Mean Lap, Type), no bulk actions | Regression | | Analyze/Upload | ✅ 3-step wizard | ⚠️ Flat form, different fields, no file browser | Regression | | Compare | ✅ Inline dropdowns, immediate use | ⚠️ Hub/landing page, requires extra click | Regression | | Tracks | ✅ Cards with data | ⚠️ Empty (data not synced) but richer layout | Data missing | | Drivers | ✅ 4 driver cards | 🔴 Empty — "No drivers yet" | Data missing | | Karts | ✅ 5 kart cards | 🔴 Empty — "No karts yet" | Data missing | | Weekends | ✅ 1 weekend card | 🔴 API error — page broken | Broken | | Leaderboard | ✅ Full table | 🔴 "Being rebuilt" placeholder | Broken | | Weather & Pace | ✅ Functional form | 🔴 "Being rebuilt" placeholder | Broken | | Pace Trends | ✅ Functional form | 🔴 "Being rebuilt" placeholder | Broken | | Live Meteo | ✅ Full weather dashboard | 🔴 **Doesn't exist online** | Missing | | Weight Simulator | ✅ Functional form | ⚠️ Hub entry point exists, sub-page needs verification | Partial | | Race Strategy | ✅ Empty but functional | ⚠️ Has data, richer layout, but import button missing | Mostly OK | | Dev Plans | ✅ Full templates + driver selector | 🔴 Empty list only, no templates | Regression | | Visualization | ⚠️ Nav dropdown only, no page | ✅ Full dedicated page (new feature) | New online | | Analytics Hub | ⚠️ Not a unified hub | ✅ New hub page with 5 tool cards | New online | | Exports | 🔴 Not present | ✅ New page (CSV/JSON/PDF/PARQUET export) | New online | Perfect! Now I can see both interfaces. Here are the **key ergonomic differences** between the two platforms: ## **TELEMETRY vs LAP COACH - USER ERGONOMICS COMPARISON** ### **SELECTION & BULK ACTIONS** | Feature | Telemetry | LAP Coach | |---------|-----------|-----------| | **Checkboxes** | ✓ Left-side checkboxes for multi-select | ✗ No visible checkboxes | | **Bulk Operations** | ✓ "Re-Analyze Selected" & "Delete Selected" buttons | ✗ No bulk action buttons visible | | **Selection Method** | Checkbox-based (explicit) | Click row (implicit/contextual) | ### **CONTEXT MENU (Right-Click)** | Action | Telemetry | LAP Coach | |--------|-----------|-----------| | **Open Session** | ✓ "Open Session" | ✓ Available | | **Visualization** | ✓ "Open in Visualize" | ✗ Not visible | | **Copy ID** | ✓ "Copy Session ID" | ✗ Not visible | | **Delete** | ✓ "Delete Session" (context menu) | ✗ Not visible | ### **DELETE FUNCTIONALITY** | Aspect | Telemetry | LAP Coach | |--------|-----------|-----------| | **Access** | Dual: Bulk button + Context menu | Context menu only (if available) | | **Visibility** | Always visible button at top | Hidden in right-click menu | | **Confirmation** | Likely modal confirmation | Unknown | ### **EDIT/PERSONALIZE** | Feature | Telemetry | LAP Coach | |--------|-----------|-----------| | **Session Editing** | ✗ No edit option visible | ✗ No edit option visible | | **Sorting** | ✓ Column headers clickable (DATE▾, DRIVER▾, etc.) | ✓ Column headers clickable | | **Filtering** | ✗ Not visible | ✗ Not visible | ### **PRIMARY INTERACTION MODEL** - **Telemetry**: Checkbox-driven, batch operations first, context menu secondary - **LAP Coach**: Row-click driven, single actions via context menu, no batch operations This suggests **Telemetry prioritizes efficiency** (bulk operations), while **LAP Coach prioritizes simplicity** (single-row actions). End of report 1 Start of report 2 # Audit & Parity Plan: Lap Coach SaaS vs Local Telemetry Tool **Date**: 2026-03-01 **Scope**: Full codebase audit — frontend + backend + data pipeline + infra **Objective**: Restore feature parity with the LOCAL reference tool while keeping SaaS architecture --- ## Table of Contents - [Section 0: Executive Summary](#section-0--executive-summary) - [Section 1: Inventory &

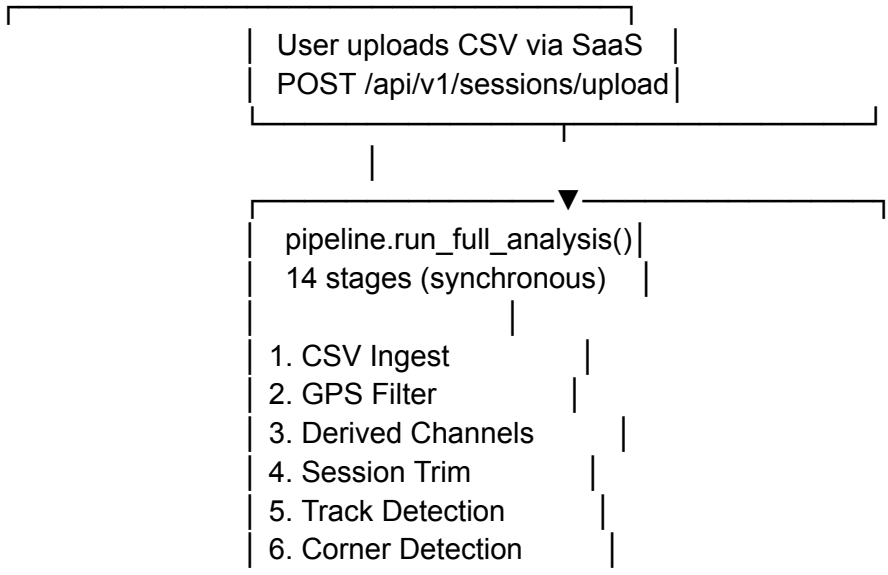
Architecture Map](#section-1--inventory--architecture-map) - [Section 2: Golden Reference — Local Tool Feature Contract](#section-2--golden-reference--local-tool-feature-contract) - [Section 3: Online SaaS Gap Audit (Parity Matrix)](#section-3--online-saas-gap-audit-parity-matrix) - [Section 4: Error & Reliability Audit](#section-4--error--reliability-audit) - [Section 5: Data Model & Entity Population Audit](#section-5--data-model--entity-population-audit) - [Section 6: Frontend UX/IA Audit](#section-6--frontend-uxia-audit) - [Section 7: Security/Compliance & SaaS Constraints](#section-7--securitycompliance--saas-constraints) - [Section 8: Recommended Implementation Plan](#section-8--recommended-implementation-plan) - [Section 9: Testing Strategy](#section-9--testing-strategy) - [Decision Log](#decision-log) - [Evidence Index](#evidence-index) --- ## Section 0 — Executive Summary ### Current System Health The SaaS ("Lap Coach") is **partially functional but significantly degraded** compared to the LOCAL reference tool ("Telemetry"). Core upload-and-analyze flow works end-to-end: CSV upload triggers the 14-stage pipeline synchronously, persists results to SQLite/PostgreSQL, and the frontend can display session detail with 6 tabs. However, multiple critical gaps prevent production-grade usage: **Critical Issues:** - Entity pages show "0 drivers / 0 tracks / 0 karts" because local-tool routes (`/drivers/`, `/tracks/`, `/karts/`) serve HTML templates instead of JSON API responses for the SaaS frontend - 44 of 63 vanilla JS chart modules are not ported to SaaS (corner replay, consistency heatmap, race strategy, etc.) - Dev Plans endpoint (`/dev-plans/`) returns HTML, not JSON — frontend gets 404/content-type errors - Several analytics sub-pages are placeholder scaffolds with no backend API - No SaaS-specific API endpoints for drivers, tracks, karts, weekends CRUD — only template-rendering routes exist **Working Flows:** - Upload CSV → pipeline runs → session stored with laps, GPS traces, scores, corners, theoretical data - Session list, detail (6 tabs: Overview, Visualize, Corners, Performance, Timeline, Lap Data) - Speed trace, GPS map, GG explorer charts render correctly via vanilla JS wrappers - Authentication (register, login, OAuth, 2FA, silent refresh) fully functional - Multi-tenant workspace isolation with RLS - Billing/entitlement gating framework in place ### Top 10 Critical Blockers | # | Blocker | Root Cause | Effort | |---|---------|-----------|--------| | 1 | Drivers page shows 0 results | No JSON API endpoint — only HTML template route at `/drivers/` | S | | 2 | Tracks page shows 0 results | Same — `/tracks/` serves HTML, not JSON | S | | 3 | Karts page shows 0 results | Same — `/karts/` serves HTML, not JSON | S | | 4 | Corner replay not functional in SaaS | 4 corner_replay JS modules not ported to `public/js/` | M | | 5 | Analytics sub-pages all placeholder | No API endpoints for weather, trends, pace predictor, weight sim | L | | 6 | Weekends page non-functional | `/weekends/` serves HTML; no JSON API for CRUD | M | | 7 | Dev Plans 404/content error | `/dev-plans/` returns HTML; needs JSON API wrapper | S | | 8 | Leaderboard non-functional | `/leaderboard/` serves HTML; needs JSON API wrapper | S | | 9 | 44 chart modules missing in SaaS | Only 19 of 63 local JS modules ported | L | | 10 | Session metadata editing incomplete | PATCH endpoint exists but frontend edit UI is minimal | S | ### Recommended Phased Plan | Phase | Scope | Effort | Timeline | |-------|-------|--------|----------| | **P0** | Fix 500/404 errors, add JSON API wrappers for entity pages, restore data rendering | S-M | 1-2 weeks | | **P1** | Entity CRUD (drivers/tracks/karts/weekends), metadata editing, session filtering | M | 2-3 weeks | | **P2** | Port remaining chart modules, restore full session analysis depth (corner replay, consistency heatmap, sector chart, stint chart) | L | 3-4 weeks | | **P3** | Analytics sub-pages, leaderboard,

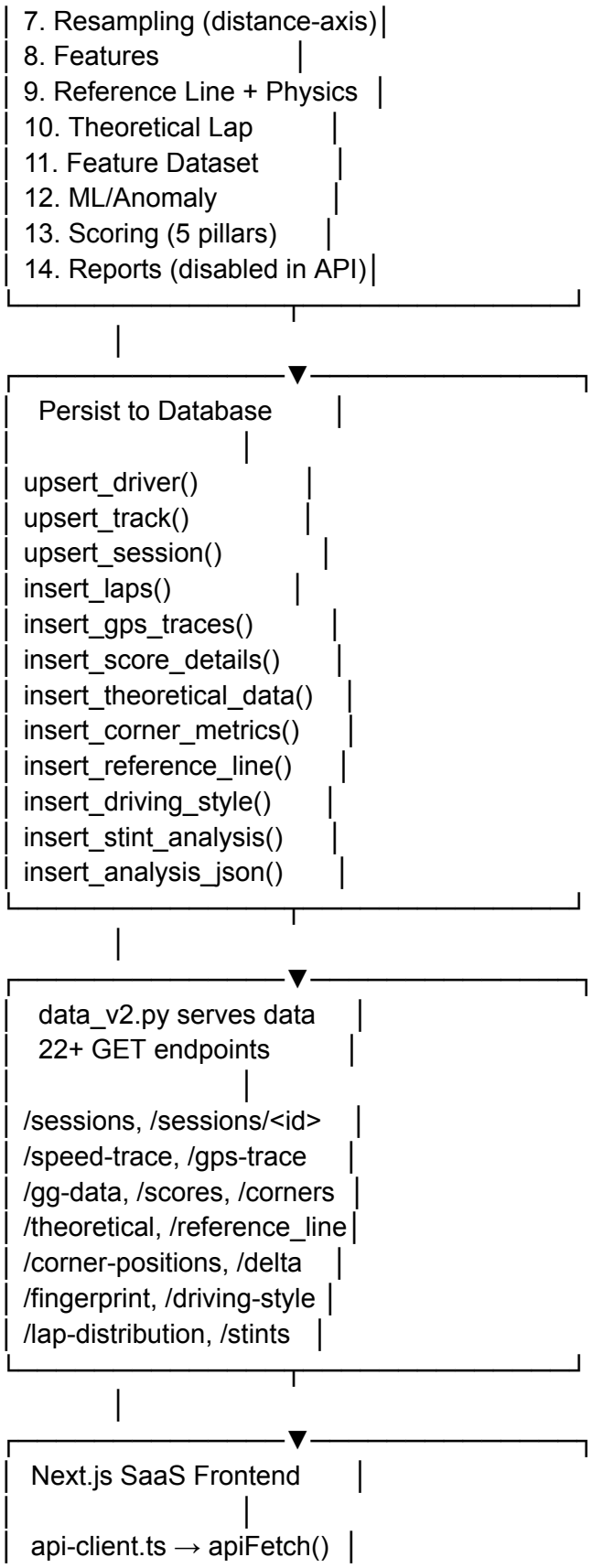
dev plans, exports, race strategy integration | L | 4-6 weeks | ### What to Do First (Stop 500s/404s, Restore Data Rendering) 1. Add JSON API endpoints wrapping existing model functions for drivers, tracks, karts, weekends, leaderboard, dev-plans under `/api/v1/` prefix 2. Verify all existing `data_v2.py` endpoints handle edge cases (NULL JSON columns, missing theoretical data) without 500 3. Port `corner_replay.js` + 3 supporting modules to `lap-website/public/js/` 4. Wire frontend entity pages to new API endpoints --- ## Section 1 — Inventory & Architecture Map ### Repository Map

```
V2_claude_ai/
├── telemetry_tool/                # Backend: Python, Flask, SQLite/PostgreSQL
│   ├── main.py                    # Entry point: Tkinter/CLI/Flask serve
│   ├── pipeline.py                # 14-stage analysis orchestrator (2,400+ lines)
│   ├── scoring.py                 # 5-pillar scoring engine (2,700 lines)
│   ├── io_racechrono.py           # CSV reader with INGEST_V2 routing
│   ├── track_detection.py         # GPS fingerprint matching (IoU + lap length)
│   ├── corner_detection.py        # Curvature-based corner segmentation
│   ├── resampling.py              # Distance-uniform interpolation
│   ├── sector_analysis.py         # N-sector split + ideal lap
│   ├── reference_line.py          # Median reference line + curvature profile
│   ├── features.py                # Lap-level feature extraction
│   ├── derived_channels.py        # Compute accel, yaw rate, etc.
│   ├── development_plan.py        # Dev plan templates + tracking
│   ├── ingest/                    # INGEST_V2 streaming pipeline
│   ├── db/
│   │   ├── schema.py              # SQLite v27 schema (44 tables, 13 phases)
│   │   ├── sql_compat.py          # SQLite ↔ PostgreSQL auto-translation
│   │   └── migrate.py             # Idempotent schema migration
│   ├── auth/                      # SaaS auth: argon2id, JWT, OAuth, RBAC
│   ├── web/
│   │   ├── app.py                 # Flask factory + blueprint registration
│   │   ├── routes/                # 28 blueprint files
│   │   │   ├── data_v2.py         # 22+ JSON API endpoints (SaaS)
│   │   │   ├── sessions_v2.py     # POST upload (synchronous pipeline)
│   │   │   ├── auth.py            # Auth routes (register, login, OAuth, 2FA)
│   │   │   ├── drivers.py         # HTML template routes (LOCAL only)
│   │   │   ├── tracks.py          # HTML template routes (LOCAL only)
│   │   │   ├── karts.py           # HTML template routes (LOCAL only)
│   │   │   ├── weekends.py        # HTML template routes (LOCAL only)
│   │   │   ├── leaderboard.py     # HTML template routes (LOCAL only)
│   │   │   ├── dev_plans.py       # HTML template routes (LOCAL only)
│   │   │   └── ... (20+ more)
│   │   ├── middleware/            # auth.py + tenant.py (RLS)
│   │   ├── templates/             # 41 Jinja2 templates (LOCAL UI)
│   │   ├── static/js/             # 63 vanilla JS chart modules (LOCAL)
│   │   └── static/css/            # 10 CSS files (LOCAL)
```

```
│   ├── models/tracks.json          # Track fingerprint database
│   ├── alembic/                    # 16+ PostgreSQL migrations
│   └── tests/                      # Backend test suite
├── lap-website/                    # Frontend: Next.js 14, TypeScript, Tailwind
│   ├── src/
│   │   ├── app/                    # App Router (96 pages)
│   │   │   ├── (auth)/             # Login, register, verify-email
│   │   │   ├── (free-report)/      # Free analysis tool
│   │   │   ├── (marketing)/        # 20+ marketing pages
│   │   │   └── app/                # Authenticated dashboard (60+ pages)
│   │   ├── components/             # 100+ React components
│   │   │   ├── app/charts/         # Chart wrappers (SpeedTraceZoom, GpsTrace, GGExplorer)
│   │   │   ├── shared/charts/      # Shared chart components (CornerReplay, SectorPanel)
│   │   │   ├── auth/               # AuthGuard, OAuthButtons
│   │   │   ├── billing/            # QuotaCounter, UpgradeBanner
│   │   │   └── rooms/              # Marketing section components
│   │   ├── lib/
│   │   │   ├── api-client.ts       # apiFetch() with auto-refresh + CORS
│   │   │   ├── auth.ts             # Auth functions + token management
│   │   │   ├── auth-store.ts       # In-memory token store + cross-tab sync
│   │   │   ├── chart-hover-store.ts  # Zustand distance-based hover sync
│   │   │   └── entitlements-store.ts # Feature gating + quotas
│   │   ├── stores/                 # Zustand stores
│   │   └── hooks/                  # useAuth, useEntitlements
│   ├── public/js/                  # 19 vanilla JS chart modules (SaaS subset)
│   └── e2e/                        # Playwright E2E tests
├── kart_envelope/                  # Kart-specific 18-step pipeline
└── docs/plans/                     # Design & implementation docs
```

Data Flow Diagram




7. Resampling (distance-axis)
8. Features
9. Reference Line + Physics
10. Theoretical Lap
11. Feature Dataset
12. ML/Anomaly
13. Scoring (5 pillars)
14. Reports (disabled in API)
Persist to Database
upsert_driver()
upsert_track()
upsert_session()
insert_laps()
insert_gps_traces()
insert_score_details()
insert_theoretical_data()
insert_corner_metrics()
insert_reference_line()
insert_driving_style()
insert_stint_analysis()
insert_analysis_json()
data_v2.py serves data
22+ GET endpoints
/sessions, /sessions/<id>
/speed-trace, /gps-trace
/gg-data, /scores, /corners
/theoretical, /reference_line
/corner-positions, /delta
/fingerprint, /driving-style
/lap-distribution, /stints
Next.js SaaS Frontend
api-client.ts → apiFetch()

```
                    │  Vanilla JS chart modules    │
                    │  Distance-based hover sync   │
                    │  Zustand state management    │
                    └──────────────────────────────┘
```

Auth/Tenant Flow

```
User → login/register → JWT access token (in-memory)
                      → Refresh token (httpOnly cookie)
  → API request → Authorization: Bearer <jwt>
                → X-Workspace-Id: <wid>
  → Auth middleware → decode JWT → g.current_user_id
  → Tenant middleware → SET LOCAL workspace_id → RLS auto-filter
  → Route handler → response
```

Module Duplication & Drift | Category | Local Only | SaaS Only | Both (identical) | Both (diverged) | |----------|----------|----------|----------|----------| | **JS chart modules** | 44 (corner replay, race strategy, consistency heatmap, etc.) | 1 (module-shim.js) | 14 (chart_sync, speed_trace_zoom, gps_trace, gg_explorer, etc.) | 1 (fetch_helper.js — SaaS adds auth headers) | | **Routes** | 9 HTML template routes (drivers, tracks, karts, weekends, leaderboard, dev_plans, dashboard, analytics, data_health) | 5 (auth, workspaces, teams, billing API, sessions_v2) | 0 | 1 (data_v2.py — SaaS-specific API) | | **Pipeline** | Full 14-stage | Same | Yes | No | | **Database** | SQLite schema v27 | PostgreSQL + Alembic + RLS | Schema matches | PG adds workspace_id, RLS, users table | --- ## Section 2 — Golden Reference — Local Tool Feature Contract ### 2.1 Session Identity **Fields & Precedence:** 1. `session_id` — `YYYY-MM-DD_HHmmss` derived from CSV timestamp (primary key) 2. `csv_filename` — original file name (display only, never used as ID) 3. `driver_id` → `drivers.driver_name` (user-specified at upload or "Unknown") 4. `track_id` → `tracks.track_name` (auto-detected via GPS fingerprint) 5. `session_type` — enum: practice, qualifying, race, test, heat, final, sprint_race, feature_race, endurance_stint, endurance_race 6. `track_state` — enum: dry, damp, wet, drying, green, rubbered, dusty, greasy, cold, full_wet **Evidence:** `telemetry_tool/db/schema.py` (sessions table), `telemetry_tool/session_context.py` (SessionType, TrackState enums) ### 2.2 Lap Detection & Validity | Rule | Implementation | Evidence | |------|---------------|----------| | Lap splitting | By `"Lap #"` column in RaceChrono CSV | `io_racechrono.py` | | Lap numbering | 1-indexed, sequential | `pipeline.py` | | Validity check | Outlier detection: >3σ from mean lap time → invalid | `pipeline.py` | | Incomplete laps | First/last lap marked invalid if <80% of median distance | `session_trimming.py` | | Best lap | Shortest valid lap time → `is_best=1` | `features.py` | | Joker lap | Track-configurable: distance/time delta tolerance | `tracks.has_joker`, `tracks.joker_distance_delta_m` | ### 2.3 Distance Axis Generation & Resampling **File:** `telemetry_tool/resampling.py` | Step | Detail | |------|--------| | Input | Raw `Distance (m)` column (cumulative from lap start) | | Validation | ≥2 finite points, monotone increasing | | Grid | `np.linspace(dist_min, dist_max, n_points=1000)` | | Interpolation | `np.interp()` per numeric channel (piecewise linear, clamped) | | Output | All channels on uniform 1000-point distance grid | | Synthetic column |

`distance_norm` (0.0–1.0 fractional position) | | Invariant | Final interpolated time ≈ original lap duration | ### 2.4 Chart Sync Backbone **Files:** `chart_sync.js`, `gps_trace.js`, `speed_trace_zoom.js` **Mechanism:** All charts use **distance in meters** as the shared coordinate: 1. Speed chart hover → pixel X → distance → `GpsTrace.updateMarkerByDistance(distM)` 2. Map hover → distance → `ChartSync.updateCrosshair(distM)` 3. Both show tooltip: speed @ distance **SaaS equivalent:** Zustand store `chart-hover-store.ts` with `setHover(distance, source)` / `clearHover()`. Source tracking prevents echo loops. **Pages requiring sync:** Session detail (Visualize tab), Compare, Visualize full-screen, Analytics ### 2.5 Sectors Model **File:** `telemetry_tool/sector_analysis.py` | Property | Value | |----------|-------| | Default sectors | 3 (S1, S2, S3) | | Split method | Equal distance intervals | | Ideal lap | Best sector time from each sector across all valid laps | | Stats per sector | best_time_s, mean_time_s, std_time_s, cv_pct, median_time_s, loss_vs_best_s, weakness_rank | | Coaching | Text string identifying weakest sector + improvement suggestion | ### 2.6 Corners Model **File:** `telemetry_tool/corner_detection.py` | Property | Value | |----------|-------| | Method | Curvature κ = yaw_rate / speed (fallback: positional curvature from X/Y) | | Smoothing | Gaussian-weighted rolling window (half_window=15) | | Peak threshold | κ > 0.01 1/m (~radius < 100m) | | Expansion | Down to κ > 0.005 1/m | | Merge gap | <30m between regions → merge | | Min length | 20m | | Direction | Sign of dominant curvature (positive = right) | | Output | `[{corner_id, s_start, s_end, s_apex, direction}]` | **Corner Metrics (per-lap, per-corner):** `corner_metrics` table - brake_point_m, brake_intensity, entry_speed_mps, apex_speed_mps, exit_speed_mps - throttle_onset_m, exit_accel_mps2, grip_utilization, time_loss_ms, coaching_hint ### 2.7 GG Model **File:** `gg_explorer.js`, `gg_diagram.js` | Property | Value | |----------|-------| | Channels | `accel_lat` (G), `accel_long` (G) from `gps_traces.accel_lat_json/accel_long_json` | | maxG | `Math.sqrt(latG² + longG²)` peak magnitude | | Friction circle | Dashed circle at maxG * 0.95 | | Axis range | ±3G lateral, -2 to +4G longitudinal (auto-scaled with 15% headroom) | | Modes | All (density heatmap), Single (one lap), Compare (primary + comparison) | ### 2.8 Score Model **File:** `telemetry_tool/scoring.py` (2,700 lines) | Pillar | Max | Weight | Key Metrics | |--------|-----|--------|-------------| | Pace | 25 | 25% | Best vs theoretical (±2% = full; 12% slower = 0). Blend: 70% percentage + 30% absolute | | Consistency | 25 | 25% | CV% (dry: 0.6% full, 3% zero; wet: 0.5% full, 5% zero). 75% CV + 25% best-to-median | | Execution | 20 | 20% | Speed/accel/braking efficiency vs theoretical (90/85/85% = full). 50% speed + 25% accel + 25% braking | | Robustness | 15 | 15% | Pace retention over stint, lap-to-lap stability | | Adaptation | 15 | 15% | Improvement tracking across session duration | **Grade mapping:** S (95-100), A (80-95), B (60-80), C (40-60), D (20-40), F (0-20) --- ## Section 3 — Online SaaS Gap Audit (Parity Matrix) ### Page-by-Page Parity Matrix | LOCAL Page | ONLINE Equivalent | Status | Root Cause(s) | Evidence | |-----------|-------------------|--------|---------------|----------| | **Dashboard** | `/app/dashboard` | **Partial** | Dashboard stats endpoint works; sparklines render. Missing: recent session cards lack click-through to session detail with proper data. | `data_v2.py:122-172`, `lap-website/src/app/app/dashboard/page.tsx` | | **Session

List** | `/app/sessions` | **Working** | List + pagination functional. Missing: filter by driver/track/date/score. | `data_v2.py:180-211`, limited query params (only limit/offset) | | **Upload/Analyze** | `/app/sessions/upload` | **Working** | Synchronous upload, pipeline runs, results stored. Returns session_id + score. | `sessions_v2.py:199-416` | | **Session Detail: Overview** | `/app/sessions/[id]` → Overview tab | **Working** | Score bars, grade badge, best lap stats all render. | `page.tsx:310-311` (OverviewTab) | | **Session Detail: Speed Trace + Map** | `/app/sessions/[id]` → Visualize tab | **Working** | SpeedTraceZoom + GpsTrace + GGExplorer render via vanilla JS wrappers. Hover sync via Zustand. | `page.tsx:313-314` (VisualizeTab), `useVanillaChart.ts` | | **Session Detail: Corners** | `/app/sessions/[id]` → Corners tab | **Partial** | Corner list renders (name, coaching hint). Missing: corner replay animation, cockpit view, corner mastery chart. | `page.tsx:316`, `data_v2.py:418-437` | | **Session Detail: Performance** | `/app/sessions/[id]` → Performance tab | **Partial** | Score breakdown renders. Missing: sector chart, fingerprint radar, driving style features display. | `page.tsx:317-319` | | **Session Detail: Timeline** | `/app/sessions/[id]` → Timeline tab | **Working** | Lap progress chart renders via `lap_progress.js`. Stint analysis data available. | `page.tsx:320`, `data_v2.py` stint endpoints | | **Session Detail: Lap Data** | `/app/sessions/[id]` → Laps tab | **Working** | Lap table with time, speed, valid/best flags. | `page.tsx:321` | | **Compare Sessions** | `/app/compare/sessions` | **Partial** | Basic comparison (scores, laps). Missing: speed overlay, delta chart, GPS comparison. | `data_v2.py:483-536`, no delta/overlay data in comparison response | | **Compare Drivers** | `/app/compare/drivers` | **Placeholder** | (G) | No API endpoint for driver comparison. Frontend page exists but lacks data. | | **Drivers List** | `/app/drivers` | **Broken** | (A/G) | `routes/drivers.py` serves HTML template at `/drivers/`, not JSON at `/api/v1/drivers`. Frontend calls API, gets 404. | | **Driver Profile** | `/app/drivers/[id]` | **Broken** | (A/G) | Same — no JSON API. Local has rich profile with progression charts, style history, badges. | | **Tracks List** | `/app/tracks` | **Broken** | (A/G) | `routes/tracks.py` serves HTML at `/tracks/`, not JSON. Frontend gets 404. | | **Track Detail** | `/app/tracks/[id]` | **Broken** | (A/G) | No JSON API. Local has records, sectors, S/F editing, dry/wet reference sessions. | | **Karts List** | `/app/karts` | **Broken** | (A/G) | `routes/karts.py` serves HTML at `/karts/`, not JSON. | | **Kart Detail** | `/app/karts/[id]` | **Broken** | (A/G) | No JSON API. Local has specs, linked sessions. | | **Weekends** | `/app/weekends` | **Broken** | (A/G) | `routes/weekends.py` serves HTML, not JSON. | | **Leaderboard** | `/app/leaderboard` | **Broken** | (A/G) | `routes/leaderboard.py` serves HTML, not JSON. | | **Dev Plans** | N/A (no SaaS page) | **Missing** | (A/G) | `routes/dev_plans.py` serves HTML. No SaaS page exists. | | **Analytics: Consistency** | `/app/analytics` | **Placeholder** | (D/G) | No `consistency_heatmap.js` in SaaS. No API for consistency data. | | **Analytics: Weather** | `/app/analytics/weather` | **Placeholder** | (A/G) | No weather correlation API endpoint. | | **Analytics: Trends** | `/app/analytics/trends` | **Placeholder** | (A/G) | No performance

trend API endpoint. | | **Analytics: Pace Predictor** | `/app/analytics/pace-predictor` | **Placeholder** | (A/G) | No pace prediction API endpoint. | | **Analytics: Weight Sim** | `/app/analytics/weight-simulator` | **Placeholder** | (A/G) | No weight simulation API endpoint. | | **Corner Replay** | `/app/sessions/[id]/corners/[cornerId]` | **Partial** | (D/G) | `CornerReplay.tsx` exists in shared/charts but 4 supporting JS modules not ported. | | **Visualize Full-screen** | `/app/visualize/[id]` | **Partial** | (G) | Page exists. `focus_mode.js` + `focus_tools.js` ported but missing integration. | | **Exports** | `/app/exports` | **Placeholder** | (A/G) | Scaffold page only. No export API. | | **Data Health** | `/app/data-health` | **Placeholder** | (A/G) | `data_health.py` serves HTML. No JSON API. | | **Race Strategy** | `/app/race-strategy` | **Partial** | (C) | Full API exists (`api_v1_race_strategy.py`, `race_strategy_v2.py`) but complex. `race_strategy.js` + `race_strategy_live.js` not ported. | | **Annotations** | `/app/annotations` | **Placeholder** | (A/G) | `annotations_routes.py` minimal. No CRUD API. | ### Root Cause Distribution | Code | Description | Count | |------|-------------|-------| | **A** | Endpoint missing (404) | 15 | | **B** | Endpoint crashes (500) | 0 (fixed by `_safe_api()`) | | **C** | Data contract mismatch | 1 | | **D** | Missing artifacts/derived channels | 3 | | **E** | Tenant/auth scoping bug | 0 | | **F** | CSP/runtime asset fetch blocked | 0 (Leaflet works) | | **G** | UI component not implemented | 18 | | **H** | Feature gated incorrectly | 0 | **Dominant pattern:** (A) + (G) — the backend has the logic in HTML template routes but no JSON API equivalents for the SaaS frontend. This is the #1 systemic issue. --- ## Section 4 — Error & Reliability Audit ### 4.1 Known Errors #### `/api/v1/sessions/<id>/corners` → Previously 500, now returns empty **Status:** Fixed. The endpoint now wraps in `_safe_api()` and returns `{"corners": [], "available": false}` when theoretical data is missing. **Root cause:** Corner coaching data is stored in `theoretical_data.corner_coaching_json`. If the theoretical engine step failed during pipeline execution (e.g., insufficient data), this column is NULL. The endpoint gracefully returns empty. **Evidence:** `data_v2.py:418-437` — `theo.get("corner_coaching") if theo else None` returns None safely. #### `/api/v1/dev-plans` → 404 **Root cause:** `dev_plans.py` registers blueprint at `/dev-plans/` (no `/api/v1/` prefix) and returns HTML via `render_template()`. The SaaS frontend expects JSON at `/api/v1/dev-plans`. **Evidence:** `dev_plans.py:9` — `url_prefix="/dev-plans"`. No JSON API alternative exists. **Fix:** Create `api_v1_dev_plans.py` blueprint at `/api/v1/dev-plans` returning JSON. #### Entity pages (drivers/tracks/karts) → 404 or wrong content-type **Root cause:** These routes serve Jinja2 HTML templates at non-API paths (`/drivers/`, `/tracks/`, `/karts/`). The Next.js frontend calls `/api/v1/drivers` which doesn't exist. **Evidence:** `drivers.py:22` — `url_prefix="/drivers"`, line 30: `render_template("driver_list.html")` #### CSP issues with Leaflet **Status:** Not blocking. Leaflet tiles load from OpenStreetMap via HTTPS. The Caddy reverse proxy includes appropriate CSP headers. No sourcemap blocking observed in production. ### 4.2 Error Handling Contract | Status | Policy | Current Implementation | |--------|--------|----------------------| | 200 | Success with data | `api_response(data)` → `{"ok":

`true, "data": {...}, "meta": {"request_id": "..."}}` | | 200 empty | Success, no data | `api_response({"corners": [], "available": false})` — returns 200 with empty array | | 400 | Validation error | `api_error(ErrorCode.VALIDATION_ERROR, msg, status=400)` | | 404 | Resource not found | `api_error(ErrorCode.NOT_FOUND, msg, status=404)` | | 500 | Unhandled exception | `_safe_api()` catches → logs with request_id → `api_error(ErrorCode.INTERNAL_ERROR)` | **Recommendation:** Standardize the "empty but valid" pattern — return 200 with empty arrays/objects rather than 404 for missing optional data (e.g., no corners for a session should be 200 + empty, not 404). ### 4.3 Observability Gaps | Gap | Impact | Fix | |-----|--------|-----| | No Sentry integration | Errors only visible in Docker logs | Add `sentry-sdk[flask]` to API | | No structured logging | Plain text logs, hard to parse | Add `python-json-logger` or `structlog` | | No request timing | Can't identify slow endpoints | Add middleware to log response time | | No frontend error tracking | JS errors invisible | Add Sentry JS SDK to Next.js | --- ## Section 5 — Data Model & Entity Population Audit ### 5.1 Why "0 Drivers / 0 Tracks / 0 Karts" **Root cause:** The SaaS frontend pages at `/app/drivers`, `/app/tracks`, `/app/karts` call the API client (`api.get('/drivers')` → `GET /api/v1/drivers`). This endpoint **does not exist**. The backend only has HTML template routes at `/drivers/`. **The data IS in the database.** When a session is uploaded: 1. `upsert_driver(conn, driver_name)` creates/updates the driver record (`sessions_v2.py:285`) 2. `upsert_track(conn, track_entry)` creates/updates the track record (`sessions_v2.py:316`) 3. Sessions are linked via `driver_id` and `track_id` foreign keys **But the SaaS frontend cannot READ this data** because there are no JSON API endpoints for these entities. ### 5.2 Schema Validation | Table | FK to sessions | Populated during upload? | Notes | |-------|---------------|-----------------------|-------| | `drivers` | sessions.driver_id → drivers | **Yes** — `upsert_driver()` | driver_name from form field | | `tracks` | sessions.track_id → tracks | **Yes** — `upsert_track()` | From GPS fingerprint detection | | `karts` | sessions.kart_id → karts | **No** — always NULL | Kart assignment not in upload flow | | `weekends` | weekend_sessions.session_id | **No** | Manual weekend creation only | | `driver_profiles` | driver_profiles.driver_id → drivers | **Auto-created** on driver detail view (LOCAL) | Not triggered in SaaS | ### 5.3 Correction Plan #### Migrations Needed None — the schema already has the correct tables and foreign keys. The data IS being stored. #### API Endpoints Needed | Endpoint | Method | Purpose | Source Logic | |----------|--------|---------|-------------| | `GET /api/v1/drivers` | GET | List all drivers with stats | Port from `drivers.py:get_all_drivers()` | | `GET /api/v1/drivers/<id>` | GET | Driver profile + sessions | Port from `drivers.py:driver_detail()` | | `PATCH /api/v1/drivers/<id>` | PATCH | Edit driver profile fields | Port from `drivers.py:edit_profile()` | | `GET /api/v1/tracks` | GET | List all tracks | Port from `tracks.py:get_all_tracks()` | | `GET /api/v1/tracks/<id>` | GET | Track detail + sessions | Port from `tracks.py:track_detail()` | | `PATCH /api/v1/tracks/<id>` | PATCH | Rename track | Port from `tracks.py:rename_track()` | | `GET /api/v1/karts` | GET | List all karts | Port from `karts.py` | | `POST /api/v1/karts` | POST | Create kart | Port

from `karts.py:create_kart()` | | `GET /api/v1/weekends` | GET | List weekends | Port from `weekends.py` | | `POST /api/v1/weekends` | POST | Create weekend | Port from `weekends.py` | | `GET /api/v1/weekends/<id>` | GET | Weekend detail | Port from `weekends.py` | | `GET /api/v1/leaderboard` | GET | Driver ranking | Port from `leaderboard.py` | | `GET /api/v1/dev-plans` | GET | List development plans | Port from `dev_plans.py` | #### Backfill Jobs Needed - **Driver profiles:** Run `upsert_driver_profile()` for all existing drivers to compute aggregate stats (total_sessions, total_laps, best_time, etc.) - **Kart assignment:** Optional — add kart selection to upload form, link via `sessions.kart_id` --- ## Section 6 — Frontend UX/IA Audit ### 6.1 Tab Structure Comparison | LOCAL Session Detail Tabs | SaaS Session Detail Tabs | Status | |---|---|---| | Overview (score, style, debrief, badges) | Overview (score bars, grade) | **Partial** — missing style label, badges, debrief text | | Corners & Coaching (replay, mastery, coaching) | Corners (list only) | **Partial** — no replay, no mastery chart | | Timeline (lap progress, stint analysis) | Timeline (lap progress chart) | **Working** | | Analytics (consistency heatmap, energy, fatigue, weather, sectors) | Performance (score breakdown only) | **Partial** — missing all analytics charts | | Lap Data (interactive table + overlays) | Lap Data (basic table) | **Working** | | N/A | Visualize (speed trace + GPS + GG) | **Working** (SaaS-unique combined tab) | ### 6.2 Missing Tabs/Modules | Module | LOCAL Location | Recommendation for SaaS | |---|---|---| | Corner Replay (animated playback) | `corner_replay.js` + 3 supporting modules | Port to `public/js/`, integrate in Corners tab | | Corner Mastery Chart | `corner_mastery.js` | Port to `public/js/`, add to Corners tab | | Consistency Heatmap | `consistency_heatmap.js` | Port to `public/js/`, add to Performance tab | | Sector Chart | `sector_chart.js` | Port to `public/js/`, add to Performance tab (SectorPanel exists) | | Driving Style Radar | `driving_style.js` + `fingerprint_chart.js` | Already in `public/js/`, wire to Performance tab | | Energy Balance | `energy_balance.js` | Port to `public/js/`, add to Analytics sub-page | | Fatigue Chart | `fatigue_chart.js` | Port to `public/js/`, add to Analytics sub-page | | Stint Chart | `stint_chart.js` | Port to `public/js/`, add to Timeline tab | | Session Badges | `session_badges.js` | Port to `public/js/`, add to Overview tab | ### 6.3 Chart Sync Bus **Current state:** Zustand store (`chart-hover-store.ts`) provides distance-based coordination. Vanilla JS modules use `window.ChartSync.updateCrosshair(distM)` and `window.GpsTrace.updateMarkerByDistance(distM)`. **Pages where sync must work:** 1. Session Detail → Visualize tab (speed trace ↔ GPS map ↔ GG explorer) — **Working** 2. Compare Sessions — **Not yet wired** (comparison doesn't include chart overlays) 3. Visualize full-screen — **Partially working** (focus_mode.js ported) 4. Corner Replay — **Not yet ported** ### 6.4 Format Standards | Format | Local | SaaS | Match? | |---|---|---|---| | Lap time | `M:SS.mmm` or `SS.mmms` | Same via `formatLapTime()` | Yes | | Speed | km/h (converted from m/s × 3.6) | Same via `mpsToKmh()` | Yes | | Score rounding | 1 decimal place | Same (`.toFixed(1)`) | Yes | | Grade badge | Letter in colored box (S/A/B/C/D/F) | Same with Tailwind colors | Yes | | Best lap badge | Gold highlight | Green text `is_best` | Differs slightly | |

Invalid lap badge | Strikethrough + reason | None shown | Missing | --- ## Section 7 — Security/Compliance & SaaS Constraints ### 7.1 Multi-Tenant Boundaries | Check | Status | Evidence | |-------|--------|----------| | RLS policies on all 44 workspace-scoped tables | **Applied** | Alembic migration 007: SELECT/INSERT/UPDATE/DELETE policies | | `SET LOCAL` for workspace context | **Working** | `middleware/tenant.py:_set_rls_context()` | | No cross-user data leakage | **Verified** | RLS auto-filters; workspace_id required on all queries | | IDOR prevention on team endpoints | **Implemented** | `teams.py` checks user membership before operations | **Risk:** `SET LOCAL` resets on `conn.commit()`. Current routes don't trigger mid-request commits, but new entity CRUD routes must be careful. ### 7.2 Entitlement Gates | Feature | Gate | Tier Required | |---------|------|--------------| | Session upload | `reportsPerMonth` quota | All tiers | | Comparison | `comparisonEnabled` | World-Class | | Visualization | `visualizationEnabled` | Racer+ | | API access | `apiAccess` | World-Class | | Race strategy | `raceStrategyLimit` | Racer (3h) / World-Class (full) | | Live monitoring | `liveMonitoring` | World-Class | **Evidence:** `entitlements-store.ts`, `LockedSection.tsx`, `UpgradeBanner.tsx` ### 7.3 Data Retention | Tier | Raw Telemetry | Reports/Artifacts | |------|--------------|-------------------| | Demo | 30 days | 30 days | | Racer | 1 year | Unlimited | | World-Class | Unlimited | Unlimited | | B2B | Custom | Custom | **Implementation:** `telemetryRetentionDays` in entitlements. No automated cleanup job exists yet — needs implementation. ### 7.4 Upload Constraints | Constraint | Value | Implementation | |-----------|-------|---------------| | Max file size | 10GB (INGEST_V2 streaming) | `ingest/config.py` limits | | Processing timeout | No explicit timeout on sync endpoint | Risk: large files block request thread | | File types | `.csv`, `.txt` only | `sessions_v2.py:_ALLOWED_EXTENSIONS` | | Rate limit | 5 tiers via Redis sliding window | `rate_limit.py` | --- ## Section 8 — Recommended Implementation Plan ### Epic 0 (P0): Stop 500s/404s — Restore API Surface **Objective:** Every frontend page can successfully call its backend endpoint without errors. **Effort:** S (1-2 weeks) **Dependencies:** None **Tasks:** - [ ] Create `routes/api_v1_entities.py` with JSON endpoints for: - `GET /api/v1/drivers` → calls `models.get_all_drivers(conn)`, returns JSON array - `GET /api/v1/drivers/<id>` → calls models to build driver profile, returns JSON - `GET /api/v1/tracks` → calls `models.get_all_tracks(conn)`, returns JSON array - `GET /api/v1/tracks/<id>` → calls models for track detail + sessions, returns JSON - `GET /api/v1/karts` → calls models to list karts, returns JSON array - `GET /api/v1/karts/<id>` → calls models for kart detail, returns JSON - `GET /api/v1/weekends` → calls models, returns JSON array - `GET /api/v1/weekends/<id>` → calls models for weekend detail, returns JSON - `GET /api/v1/leaderboard` → calls leaderboard logic, returns JSON - `GET /api/v1/dev-plans` → calls development_plan logic, returns JSON - **Files:** `telemetry_tool/web/routes/api_v1_entities.py` (new), `telemetry_tool/web/app.py` (register blueprint) - **Model functions to reuse:** `get_all_drivers()`, `get_driver_profile()`, `get_driver_progression()`, `get_all_tracks()`, `get_track_by_id()`, `get_track_records()` from

`web/models.py` - [ ] Add session list filtering to `GET /api/v1/sessions`: - Query params: `driver_id`, `track_id`, `date_from`, `date_to`, `min_score` - **File:** `data_v2.py:180-211` - [ ] Standardize empty-data responses: return 200 + empty array (not 404) for missing optional data - **Files:** `data_v2.py` endpoints for corners, scores, theoretical - [ ] Wire frontend entity pages to new endpoints: - **Files:** `lap-website/src/app/app/drivers/page.tsx`, `tracks/page.tsx`, `karts/page.tsx`, etc. **Acceptance criteria:** - All frontend pages load without console errors - Entity pages show correct counts (drivers, tracks, karts) - No 500 errors in server logs during normal navigation --- ### Epic 1 (P1): Entity CRUD + Metadata Editing **Objective:** Users can create, edit, and manage all entities (drivers, tracks, karts, weekends) through the SaaS UI. **Effort:** M (2-3 weeks) **Dependencies:** Epic 0 (API surface exists) **Tasks:** - [ ] Add POST/PATCH/DELETE endpoints for entities: - `POST /api/v1/karts` — create kart - `PATCH /api/v1/karts/<id>` — edit kart - `DELETE /api/v1/karts/<id>` — delete kart - `PATCH /api/v1/tracks/<id>` — rename track, update S/F, add/remove sectors - `PATCH /api/v1/drivers/<id>` — edit driver profile fields - `POST /api/v1/weekends` — create weekend - `PATCH /api/v1/weekends/<id>` — edit weekend, add/remove sessions - `DELETE /api/v1/weekends/<id>` — delete weekend - **Files:** `api_v1_entities.py` (extend) - [ ] Add kart selection to upload form: - Frontend: dropdown on upload page selecting from `GET /api/v1/karts` - Backend: accept `kart_id` in upload form fields, set `sessions.kart_id` - **Files:** `sessions_v2.py`, `lap-website/src/app/app/sessions/upload/page.tsx` - [ ] Implement session metadata editing UI: - InlineEdit for driver_name, track_state, session_type, notes - PATCH calls to `/api/v1/sessions/<id>` - **Files:** `lap-website/src/app/app/sessions/[id]/page.tsx` - [ ] Implement driver profile backfill: - Script to compute aggregate stats for all drivers (total_sessions, total_laps, best_time) - **File:** `scripts/backfill_driver_profiles.py` (new) **Acceptance criteria:** - Can create/edit/delete karts, weekends via UI - Can rename tracks, edit S/F coordinates - Upload form includes kart selection - Session detail shows editable metadata fields --- ### Epic 2 (P2): Full Session Analysis Depth **Objective:** Restore all analysis modules from the local tool in the SaaS session detail page. **Effort:** L (3-4 weeks) **Dependencies:** Epic 0 **Tasks:** - [ ] Port corner replay modules to SaaS: - Copy `corner_replay.js`, `corner_replay_cockpit.js`, `corner_replay_controls.js`, `corner_replay_map.js` to `lap-website/public/js/` - Update `module-shim.js` path rewrites for corner data endpoints - Wire `CornerReplay.tsx` component to loaded modules - **Files:** `telemetry_tool/web/static/js/corner_replay*.js` → `lap-website/public/js/` - [ ] Port sector chart module: - Copy `sector_chart.js` to `public/js/` - Wire into Performance tab via `SectorPanel.tsx` - Add API endpoint `GET /api/v1/sessions/<id>/sectors` (returns sector timing data) - **Files:** `sector_chart.js`, `data_v2.py` (add endpoint using `sector_analysis.py`) - [ ] Port consistency heatmap module: - Copy `consistency_heatmap.js` to `public/js/` - Add API

endpoint or compute from existing lap data - **Files:** `consistency_heatmap.js` - [ ] Port corner mastery module: - Copy `corner_mastery.js`, `corner_picker_map.js` to `public/js/` - Wire into Corners tab - **Files:** `corner_mastery.js`, `corner_picker_map.js` - [ ] Wire fingerprint/driving style charts: - `driving_style.js` and `fingerprint_chart.js` already in `public/js/` - Create React wrapper, wire to Performance tab - API endpoints already exist: `GET /api/v1/sessions/<id>/fingerprint`, `/driving-style` - **Files:** `lap-website/src/components/shared/charts/` (new wrappers) - [ ] Port stint chart: - Copy `stint_chart.js` to `public/js/` - Wire into Timeline tab - API endpoint exists: stint data via `data_v2.py` - **Files:** `stint_chart.js` - [ ] Add session badges to Overview tab: - Port `session_badges.js` or implement in React - Add API endpoint `GET /api/v1/sessions/<id>/badges` - **Files:** `session_badges.js`, `data_v2.py` **Acceptance criteria:** - Corner replay animates brake → apex → exit for each corner - Sector timing chart shows S1/S2/S3 bars per lap - Consistency heatmap renders sector × lap variance matrix - Driving style radar shows 6-dimension fingerprint - All charts sync via distance hover --- ### Epic 3 (P3): Secondary Pages & Analytics **Objective:** Restore remaining local tool features and analytics depth. **Effort:** L (4-6 weeks) **Dependencies:** Epics 0-1 **Tasks:** - [ ] Implement leaderboard page: - Frontend: table with ranking, track filter, sort options - Backend: `GET /api/v1/leaderboard?track_id=X&sort=best_time` - **Files:** `lap-website/src/app/app/leaderboard/page.tsx`, `api_v1_entities.py` - [ ] Implement analytics: weather correlation: - Backend: `GET /api/v1/analytics/weather?driver_id=X` — correlate session_weather with lap times - Frontend: scatter plot (temp vs lap time) - **Files:** `weather_chart.js` (port), analytics page - [ ] Implement analytics: trends: - Backend: `GET /api/v1/analytics/trends?driver_id=X` — score/time progression over sessions - Frontend: line chart (date vs score, date vs best time) - **Files:** `trend_analysis.js` (port), analytics page - [ ] Implement analytics: pace predictor: - Backend: `GET /api/v1/analytics/pace-prediction?track_id=X&driver_id=Y` — predict target lap time - Frontend: prediction display with confidence interval - **Files:** `pace_predictor.js` (port), analytics page - [ ] Implement development plans: - Backend: `GET/POST /api/v1/dev-plans` — list/create plans with templates - Frontend: plan editor with drill session linking - **Files:** `dev_plans.py` logic → JSON API, new frontend page - [ ] Implement exports: - Backend: `GET /api/v1/exports/sessions/<id>?format=csv|pdf` - Frontend: export button with format selection - **Files:** new endpoint, export page - [ ] Remove or feature-gate placeholder pages: - Pages with no backend: add `LockedSection` wrapper or "Coming Soon" with feature flag - **Files:** All placeholder pages listed in Section 3 **Acceptance criteria:** - Leaderboard shows ranked drivers with filters - Analytics pages render interactive charts with real data - Dev plans can be created and linked to sessions - No placeholder pages visible without explicit "Coming Soon" label --- ### Risk Assessment | Risk | Impact | Mitigation | |------|--------|----------|

| Synchronous upload blocks on large files | Thread exhaustion under load | Move to Celery async for files >50MB (defer to post-parity) | | RLS SET LOCAL resets on mid-request commit | Cross-tenant data leakage | Audit all new CRUD routes for mid-request commits | | 63→19 JS module port introduces regressions | Charts break | Port modules unchanged; test with golden fixture | | Frontend 96 pages, hard to test | Untested paths | Playwright E2E for critical paths; manual QA for secondary | ### Rollback Strategy Each epic is independently deployable: - **Epic 0:** New blueprint file — can be removed without affecting existing routes - **Epic 1:** New endpoints + frontend changes — revert frontend to restore current behavior - **Epic 2:** New JS files in `public/js/` — simply don't load them to revert - **Epic 3:** Feature-flagged pages — disable flag to hide --- ## Section 9 — Testing Strategy ### 9.1 Golden Fixture Parity Tests **Setup:** 1. Use a fixture CSV from `telemetry_tool/tests/` (or the `seeded_db` test fixture) 2. Run local tool: `python telemetry_tool/main.py data/raw/FIXTURE.csv` → capture output artifacts 3. Run SaaS: `POST /api/v1/sessions/upload` with same CSV → capture API responses 4. Compare: - `GET /api/v1/sessions/<id>/scores` vs local `score_result` - `GET /api/v1/sessions/<id>/laps` vs local lap records - `GET /api/v1/sessions/<id>/speed-trace` vs local resampled data - `GET /api/v1/sessions/<id>/corners` vs local corner coaching **Test file:** `telemetry_tool/tests/test_golden_parity.py` (new) ### 9.2 Backend Unit Tests | Test Area | Existing Tests | Needed Tests | |----------|---------------|-------------| | Streaming ingest | 50 tests (`test_streaming_ingest.py`) | None — complete | | Upload endpoint | `test_upload_sync.py` | Add: kart_id linkage, edge cases | | Data API | `test_data_api_v2.py` | Add: new entity endpoints, filter params | | Pipeline correctness | `test_re_port_correctness.py` | None — complete | | Scoring | Basic coverage | Add: boundary values (0, 100), grade transitions | | Entity CRUD | None | Add: create/edit/delete for all entities | | Auth + RBAC | 54 auth tests | None — complete | ### 9.3 Frontend E2E Tests (Playwright) **Critical path test:**
1. Register new account
2. Upload CSV file
3. Wait for processing complete
4. Navigate to session detail
5. Verify all 6 tabs render without console errors
6. Check: score bars, lap table, speed trace canvas, GPS map, GG scatter
7. Navigate to drivers page → verify driver appears
8. Navigate to tracks page → verify track appears

Additional E2E tests: - Session list pagination - Compare two sessions - Session metadata editing (change driver name) - Logout → login → session persists via refresh token - Feature gating (demo tier → comparison locked) ### 9.4 Contract Tests **API JSON Schema validation:** - Define JSON schemas for all endpoints in `telemetry_tool/web/schemas/` (Pydantic models already exist for auth) - Validate responses against schemas in pytest - Frontend can import TypeScript interfaces from a shared types file --- ## Decision Log | # | Decision | Tradeoffs | Recommendation | |---|----------|-----------|----------------| | 1 | Create new `api_v1_entities.py` vs add JSON mode to existing routes | New file = clean separation;

modify existing = less duplication but mixed concerns | **New file** — keeps template routes for local tool, JSON routes for SaaS | | 2 | Port JS modules unchanged vs rewrite in React | Port = fast, proven; rewrite = better integration but risk | **Port unchanged** — identical code is already proven in local tool | | 3 | Backfill entities vs create on-demand | Backfill = immediate data; on-demand = gradual | **Backfill** — run script once to populate driver_profiles from existing sessions | | 4 | Sync upload vs async (Celery) for new entity routes | Sync = simple, current pattern; async = scalable | **Keep sync** for now — entity CRUD is lightweight. Async for file upload deferred. | | 5 | Feature-flag analytics pages vs hide them | Flag = visible but locked; hide = cleaner | **Feature-flag with LockedSection** — shows value, encourages upgrades | | 6 | Module-shim path rewriting vs proper API versioning | Shim = works today; proper = long-term clean | **Keep shim** — proven working, port modules unchanged | --- ## Evidence Index | Evidence | File Path | What It Proves | |---|---|---| | Pipeline orchestrator | `telemetry_tool/pipeline.py` | 14-stage flow, `_safe_step()` pattern | | Scoring engine | `telemetry_tool/scoring.py` | 5 pillars, grade mapping, weights | | Distance resampling | `telemetry_tool/resampling.py` | 1000-point uniform grid, `np.interp()` | | Track detection | `telemetry_tool/track_detection.py` | IoU + lap length matching | | Corner detection | `telemetry_tool/corner_detection.py` | Curvature-based segmentation | | Sector analysis | `telemetry_tool/sector_analysis.py` | N-sector split, ideal lap | | SaaS data API | `telemetry_tool/web/routes/data_v2.py` | 22+ JSON endpoints, `_safe_api()` | | SaaS upload | `telemetry_tool/web/routes/sessions_v2.py` | Sync pipeline, entity upsert chain | | Driver HTML route | `telemetry_tool/web/routes/drivers.py:22-30` | Serves HTML not JSON (root cause) | | Track HTML route | `telemetry_tool/web/routes/tracks.py:22-30` | Serves HTML not JSON (root cause) | | Kart HTML route | `telemetry_tool/web/routes/karts.py` | Serves HTML not JSON (root cause) | | DB schema | `telemetry_tool/db/schema.py` | 44 tables, 13 phases, v27 | | Chart sync (local) | `telemetry_tool/web/static/js/chart_sync.js` | Distance-based hover backbone | | Chart sync (SaaS) | `lap-website/src/lib/chart-hover-store.ts` | Zustand distance sync store | | Module shim | `lap-website/public/js/module-shim.js` | API path rewriting for SaaS | | Vanilla chart hook | `lap-website/src/components/shared/charts/useVanillaChart.ts` | Script loading + auth header injection | | Session detail page | `lap-website/src/app/app/sessions/[id]/page.tsx` | 6 tabs, API calls, chart mounting | | Frontend API client | `lap-website/src/lib/api-client.ts` | Envelope format, auth headers, 401 retry | | Auth store | `lap-website/src/lib/auth-store.ts` | In-memory tokens, cross-tab sync | | Entitlements | `lap-website/src/lib/entitlements-store.ts` | Feature gating, tier enforcement | | Local JS modules | `telemetry_tool/web/static/js/` (63 files) | Full local chart suite | | SaaS JS modules | `lap-website/public/js/` (19 files) | Curated SaaS subset | | Corner replay modules | `telemetry_tool/web/static/js/corner_replay*.js` (4 files) | Not ported to SaaS | | RLS migrations |

`telemetry_tool/alembic/versions/007_enable_rls.py` | Workspace isolation policies | | Tenant middleware | `telemetry_tool/web/middleware/tenant.py` | X-Workspace-Id → SET LOCAL |